

Predicting Formal Protest Petitions Against Housing Development

Evidence from Austin, Texas

Daniel Hardesty Lewis

M.S. Candidate, Urban Planning

Graduate School of Architecture, Planning and Preservation

Columbia University

April 16, 2026

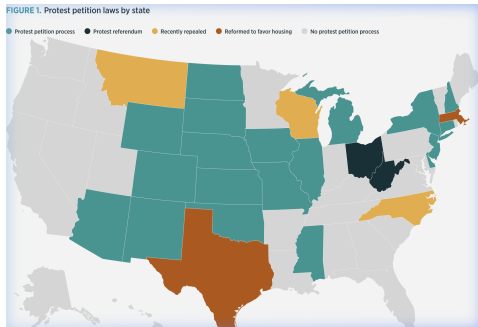
Housing Crisis and Local Opposition

The Affordability Problem

- Austin grew $\approx 40\%$ between 2010–2024.
- New housing requires discretionary rezoning approval.
- Projects face public hearings and organized opposition.

The Political Mechanism

Texas law gave adjacent property owners a formal institutional veto: the **protest petition**.

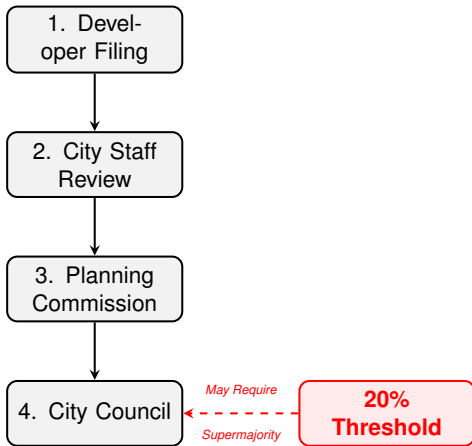


The Protest Petition: An Institutional Veto

Texas LGC Chapter 211: The 20% Rule

- Owners of 20% of land within 200ft may file written protest.
- A valid petition triggers a supermajority ($\frac{3}{4}$) council vote.
- Post-2014 council (11 members): 6 votes → 9 required.
- 21 states maintain similar petition statutes.

A localized minority can block citywide housing decisions.

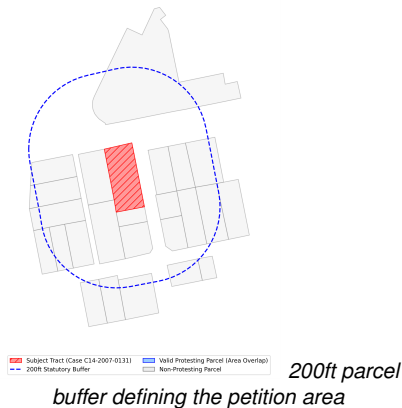


Protest Petition Rules Across States

State	Threshold	Buffer	Supermajority
Texas	20%	200ft	3/4
Iowa	20%	200ft	3/4
New York	20%	100ft	3/4
Colorado	20%	100ft	2/3
Oklahoma	50%	300ft	3/5 or 3/4
Massachusetts	50%	300ft	2/3

Source: Furth & McKinley (2025). 21 states maintain active protest petition statutes.

Visualizing the 200ft Statutory Protest Buffer
(Empirical TCAD Parcel Boundaries for Case C14-2007-0131)



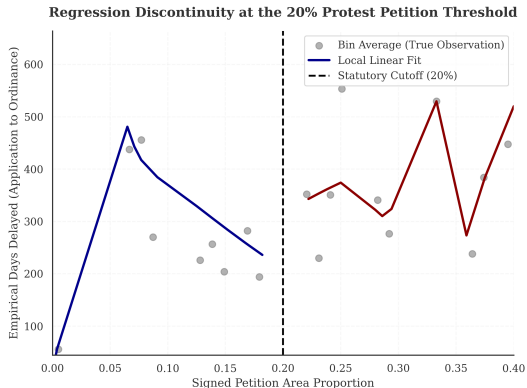
Why Petitions Matter: Institutional Costs

Reduced-Form Discontinuity at the 20% Threshold

- Running variable: reconstructed petition-share.
- Outcome: council processing time.
- Estimator: triangular-kernel WLS at the 20% margin.

Result: Crossing the threshold adds **42.5 days** (6.1 weeks).

95% CI: [26.4, 58.6] days. McCrary $p > 0.05$.



The Research Question

The Prediction Problem

Can we anticipate protest petitions *before* public hearings using only filing-date records?

Filing-Date Features

- Site geometry: lot area, height/FAR deltas.
- Current vs. requested zoning class.
- Buffer demographics and TCAD appraisals.
- Historical petition rates in the district.

A Sample Case

Feature	Case C14-2018-0156
Gross Site Area	4.92 Acres
Height Request	+15.5 ft
FAR Request	+0.43
Neighborhood	Riverside
Result	Valid Petition

Race/ethnicity excluded from predictive features; reserved for post-hoc equity audits only.

The Rarity Problem

Sample Characteristics

- $N = 6783$ cases, 2007–2024.
- Base rate: $\approx 2.6\%$ (1 in 38 cases).
- Test-set positives: $n_{\text{pos}} = 15$.

With so few events, confidence intervals are wide. The model output is a **score per case** (0–1); cases are ranked by that score, not classified by threshold.

Evaluation Design

- Train on years $\leq t$; evaluate on years $> t$.
- Temporal holdout for the test set; random cross-validation within training.

Primary metric: PR-AUC

AUROC omitted: we are predicting a rare class.

Timeline Integrity: As-Of Features

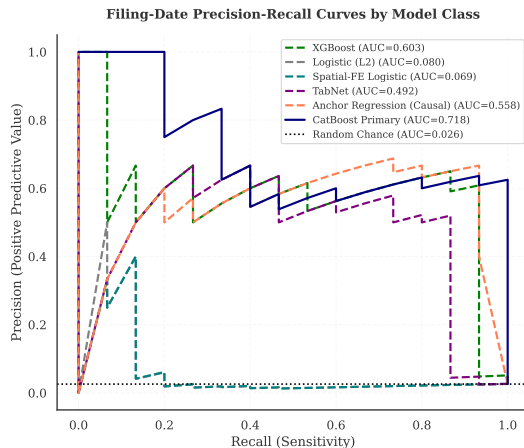
Enforcing the Constraint

- 17 years of zoning records from Austin's open-data portal (2007–2024).
- Filing dates back-calculated from statutory minimums.
- American Community Survey (ACS) and Travis County Appraisal District (TCAD) records anchored to publication date.
- $N = 8$ IRB-approved interviews (ACY0820) for institutional grounding.

COLLEGE UNIVERSITY <i>Graduate School of Arts, Sciences, Planning and Transportation</i>	Global Mobility Center <i>Study, Mobility, Global Planning</i>
Research Project Overview <i>Predicting NCCBYD use in Austin's Rising Rental Era</i>	
Project Abstract	
This research identifies opportunities in development using AI and improves the understand of that opportunity in Austin, Texas. We analyze public policy, public behavior, public housing, and existing opportunities. The study examines the demographic implications of using predictive analytics in anticipating public behavior. The project explores predictive modeling, statistical analysis, and qualitative analysis to provide an effective behavior. Study by examining of market opportunities in Austin, and consider their implications with planning solutions.	
Study Objectives	
Identify Patterns: Examine the impact of historic and geographic data in predicting patterns in transportation and study demographic implications of the opportunity.	
Examine Patterns: Analyze historic data and use predictive models to anticipate public behavior and assess opportunities in specific development projects.	
Demographic Implications: Examine how demographic trends in the implications of using algorithms to track housing policy and identifying the ethical risks of such technology.	
Participant Role	
We are seeking experienced urban planners, developers, high school mathematics teachers, and community organizers.	
Activity: A 100-hour course includes a completed study in the program.	
Work: You experience with learning, developing planning skills, and community engagement in Austin.	
Data Privacy and Ethical Standards	
This study has been approved by the College University Institutional Review Board under Protocol 101-1000.	
Participants are invited voluntarily.	
You may decline to answer any specific question or discontinue the interview at any time.	
Identifying will help identify questions in meeting public information needs from specific participants in the research.	

Predicting Resistance: Filing-Date Results

Tree Ensembles Dominate at the Top of the Risk Ranking



Operational Utility: Triage in Practice

Risk Ranking Performance

- Highest-risk 10% of cases: **25.4%** precision.
- **9.9x** lift over the base rate.
- Captures 15 true positives in 59 flagged cases.

Reviewing flagged cases is **nearly 10x** more likely to surface a genuine petition case.

Appropriate Uses

- ✓ Prioritize outreach for high-risk cases.
- ✓ Flag cases for early negotiation.
- × Hard probability cutoffs.

Four-Family Modeling Benchmark

Architecture Families Benchmarked

Family		Key Property
Tree Boost)	(Cat-	Best tabular OOD PR-AUC
Deep Net/MLP)	(Tab-	Requires L2; overfits at low N
Linear (L2 LR)		Strong at T+0; decays over time
Robust REx)	(V-	Distributional robustness base-line

Selection Criterion

All families tuned to maximize out-of-distribution PR-AUC under strict temporal separation.

Primary model (CatBoost) selected before examining test-set results: best OOD discrimination with the most stable calibration.

What Drives Petition Risk

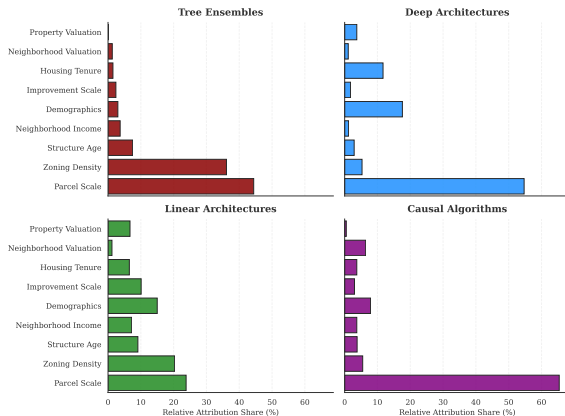
Key Finding

Tree ensembles concentrate weight on **neighborhood context**: valuation, tenure, and demographic composition.

Deep models shift weight toward **physical parcel attributes**: scale, improvement size, and geometry.

Same data, same outcome;
different internal logic.

Comparative Primary Reliance: Top Feature Clusters Across Architectures

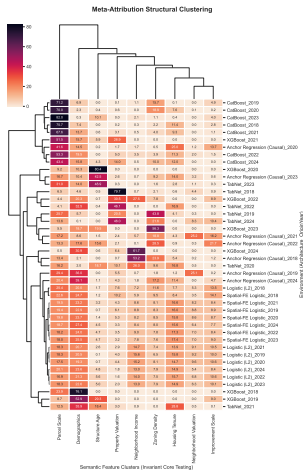


Architecture Determines Feature Reliance

Architecture-Dependent Reliance

- Trees: >80% weight on socio-demographic clusters.
- Deep models: shift toward physical and geographic attributes.
- Same data, same task: different internal logic.

Different architectures may be tapping into different mechanisms of residential opposition: neighborhood-sorting logic vs. physical disruption logic.



Temporal Drift and Neighborhood Agency

When Does the Model Break?

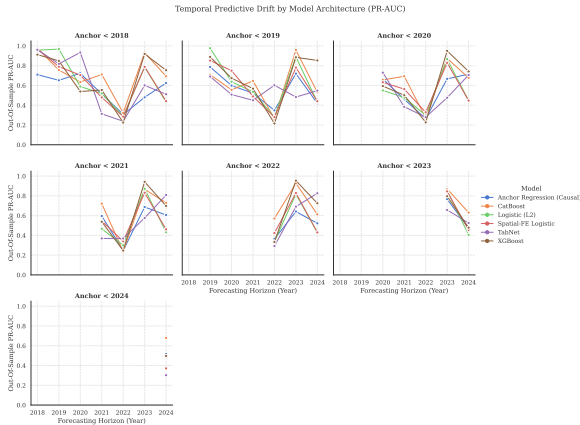
- Linear models collapse at T+3 horizons.
- Tree ensembles degrade more gracefully.

The 2022 Dip

- PR-AUC drops sharply across all families in 2022.
- Consistent with pandemic-era filing disruptions reaching the evaluation window.

Performance sensitivity to ambient shocks is *consistent with* mobilization as a strategic social choice.

Note: descriptive out-of-sample variation; no single institutional break is identified.

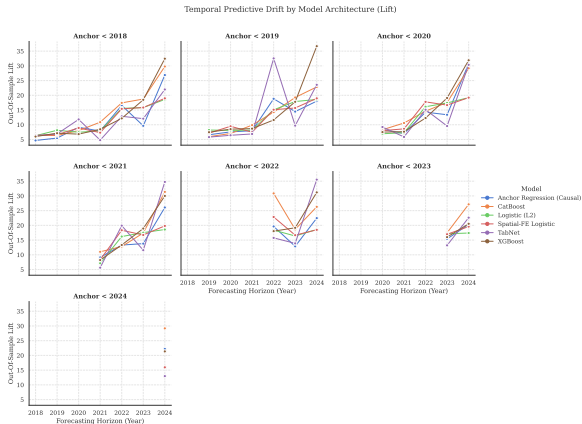


Signal Dilution: When Retraining Hurts

A Counter-Intuitive Finding

- Pre-2018 models achieve $\sim 34\times$ lift in 2024.
- Retraining on modern data degrades lift to $\sim 19\times$.

Post-ordinance labels starve the model of positive examples. The frozen legacy model preserves spatial sorting intelligence from an active institutional era.



What the Algorithm Cannot See

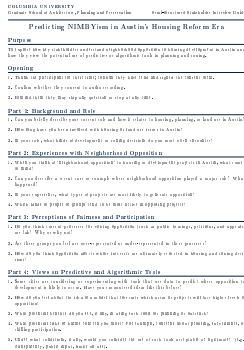
What the Algorithm Cannot Distinguish

- Exclusionary NIMBYism (affluent West Austin blocking density).
- Anti-displacement resistance (East Austin protecting tenants).

Same risk weight; opposite social meaning.

Practitioner Perspectives (IRB ACYY0820, $N = 8$)

- *“Haters are overrepresented and silent supporters are underrepresented.”* (Beaudet) → explains the model’s performance ceiling.
- Rigid cutoffs could miss communities facing real displacement risk. (Torrado) → supports triage ranking over hard thresholds.
- Formal opposition data lacks anonymous channels; it is not representative of the full community. (Tomko) → the training labels are



IRB-Approved Interview Protocol

Limitations

Methodological

- **Chilling effects:** Only filed cases observed; pre-emptive withdrawals are invisible. Selection correction was not feasible.
- **Timestamp imputation:** Milestone dates approximated from statutory minimums.
- **Unobserved certification:** Outcome is reconstructed area share, not the clerk's legal determination.

Ethical and Governance

- **Proxy discrimination:** Neighborhood income and demographics encode class and race dynamics even without protected-class inputs.
- **Motive blindness:** Cannot separate exclusionary NIMBYism from anti-displacement resistance.
- **External validity:** Parameters are Austin-specific and pre-HB 24 only.

Conclusion

What we showed

- Crossing 20% adds 42.5 days of processing time: the outcome is worth predicting.
- Filing-date data ranks petition risk above chance (PR-AUC = 0.718, CI: [0.48, 0.91]), with top-decile lift of 9.9x.
- Algorithm choice determines which feature clusters drive predictions, a normative decision.

What surprised us

- Retraining on modern data degrades lift; legacy models preserve spatial sorting intelligence.
- Performance ceilings are consistent with mobilization as a strategic social choice.

What it implies

- Predictions belong in relative triage rankings, not case-level probability thresholds.
- Methodology is portable; Austin-specific parameters are not.

Implications: A Now-Closed Policy Window

The HB 24 Shift (September 1, 2025)

- Texas HB 24 exempted residential upzonings from the supermajority requirement.
- The 20% institutional veto no longer applies to the most contested case types.
- The institutional regime is closed; its demographic and spatial correlates of opposition remain visible in the predictions.

What travels forward

- Methodology (time-stamped features, temporal holdouts, semantic clustering, qualitative integration) is portable to other discretionary review systems.
- New York, Iowa, Oklahoma, and Arizona retain active petition statutes. The framework applies wherever institutional veto mechanisms create predictable opposition patterns.

Discussion / Q&A

Thank you for your time and feedback.

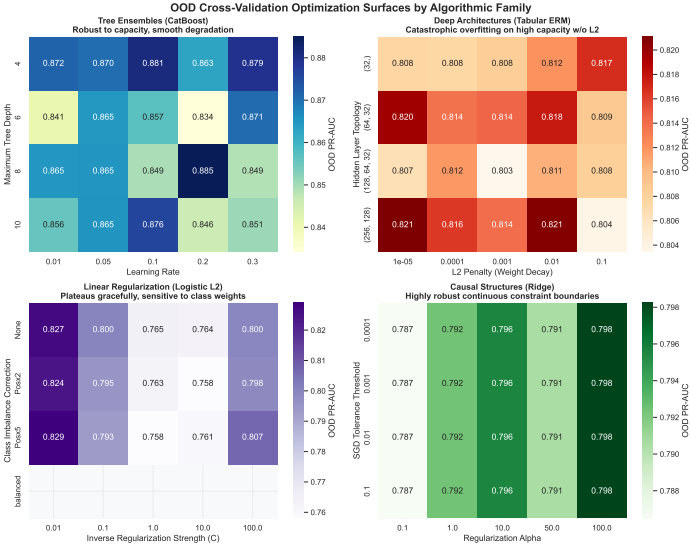
Dory Kornfeld
Weiping Wu
James Piacentini
Josh Begley

Stijn Van Nieuwerburgh
Matthew Bauer
Adam Vosburgh

Selected References I

- Furth, S., & McKinley, K. (2025). *Protest petition statutes across U.S. states*. Mercatus Center.
- Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A.V., & Gulin, A. (2018). CatBoost: Unbiased boosting with categorical features. *NeurIPS*.
- Arik, S.O., & Pfister, T. (2021). TabNet: Attentive interpretable tabular learning. *AAAI*.
- Krueger, D., Caballero, E., Jacobsen, J.H., Zhang, A., Binas, J., Zhang, D., Le Priol, R., & Courville, A. (2021). Out-of-distribution generalization via risk extrapolation (V-REx). *ICML*.
- Lundberg, S.M., & Lee, S.I. (2017). A unified approach to interpreting model predictions. *NeurIPS*.
- Imbens, G., & Lemieux, T. (2008). Regression discontinuity designs: A guide to practice. *Journal of Econometrics*.
- Texas Local Government Code §211.006 (protest petition statute).
- Texas HB 24 (2025, 89th Legislature). An Act Relating to Zoning Authority.

Appendix: Hyperparameter Robustness



Appendix: Sample Selection and Coverage

Sample Selection

Selection Step	Cases
Raw Austin open-data records	15,238
Discretionary cases only	7,153
Usable filing-year information	6,783
Analytic sample	6,783

Illustrative Panel Structure

Case	Year	Lag Pet.	Med. Inc.	Own. Sh.
C14-2015-001	2015	3	\$85k	62%
C14-2015-002	2015	0	\$42k	21%
C14-2015-006	2015	7	\$112k	85%
C14-2016-009	2016	1	\$55k	34%
C14-2016-012	2016	4	\$95k	71%

Appendix: Semantic Feature Cluster Mapping

Cluster	Representative Variables
Demographics	ACS % White, % Hispanic, % Black, % Asian
Housing Tenure	ACS owner-occupied, renter-occupied, total units
Income & Rent	Median household income, gross rent, poverty count
Property Valuation	Tax assessor appraised value, market value, land value
Structure Age	Tax assessor year built, COA year built
Parcel Scale	Acres, lot size (sqft), gross area, deed acreage
Historical Protest Activity	Prior protest flag, 1-year and 3-year spatial contagion